

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

B.Tech (Electronics and Telecommunication Engineering) SEMESTER - 5 Summer 2025 (Supply.)

Course : Bachelor of Technology (Electronics and Telecommunication Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTETC502 - DIGITAL SIGNAL PROCESSING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

1. What is the minimum sampling rate required to perfectly reconstruct a signal with maximum frequency component of 8 kHz?
A) 16 kHz
B) 8 kHz
C) 4 kHz
D) 32 kHz
2. Which property of the DFT states that circular convolution in time domain corresponds to multiplication in frequency domain?
A) Linearity
B) Time Shifting
C) Modulation
D) Convolution
3. The Z-transform can be derived from the Laplace transform by which of the following substitution?
A) $z = sT$
B) $s = zT$
C) $z = e^{sT}$
D) $s = e^{zT}$
4. An LTI system is **causal and stable** if the ROC of its Z-transform:
A) Includes the origin
B) Includes the entire Z-plane
C) Includes the unit circle and lies outside the outermost pole
D) Is inside the innermost pole
5. In power series method, the solution of a system is represented as:
A) Infinite series in powers of n or z
B) Polynomial of finite degree
C) Sum of sinusoids
D) Sum of exponentials
6. Impulse invariance method maps:
A) Discrete-time poles to continuous-time poles
B) Continuous-time poles to discrete-time poles
C) Both poles and zeros identically
D) None of the above

- 7 The direct form realization of FIR filters primarily uses:
 A) Feedback loops
 B) Delay elements and multipliers only
 C) Differential equations
 D) Pole-zero placement
- 8 In multirate DSP, which of the following operations involves changing the sampling rate?
 A) DFT
 B) Convolution
 C) Upsampling and downsampling
 D) FFT
- 9 Downsampling a signal by a factor of 2 results in:
 A) Doubling the time resolution
 B) Halving the time resolution
 C) Increasing frequency resolution
 D) Aliasing if prefiltering is not applied
- 10 In a perfect reconstruction two-channel filter bank, the output signal is:
 A) Amplified version of input
 B) Time-reversed version of input
 C) A delayed version of input
 D) A distorted version of input
- 11 Why are ideal filters **not realizable** in practice?
 A) They require feedback
 B) They are non-causal and have infinite impulse response
 C) Their impulse response is finite
 D) They have nonlinear phase
- 12 The Z-transform is particularly useful for analyzing:
 A) Non-linear systems
 B) Discrete-time LTI systems
 C) Continuous-time signals
 D) Analog filters

Q2. Solve the following.

- A) State and prove sampling theorem with neat diagrams. 6
- B) State advantages of digital signal processing over analog signal processing. 6

Q3. Solve the following.

- A) 1. Find the inverse Fourier transform of $H(e^{j\omega}) = \delta(\omega) \quad -\pi < \omega \leq \pi$ 6
 2. Find Circular convolution of the following sequences:
 $x_1(n) = \{1, 2, 3, 4\}$, $x_2(n) = \{1, 1, 2, 2\}$ for $n=0,1,2,3$
- B) Find the impulse response of the system described by the difference equation 6
 $y(n) - 3y(n-1) - y(n-2) = x(n) + 2x(n-1)$ using z-transform.

Q4. Solve Any Two of the following.

- A) Find Z-transform and plot the respective ROCs 6
 1. $0.5^n u(n) + 2^n u(-n-1)$
 2. $-0.5^n u(-n-1) + 2^n u(-n-1)$
- B) Find the expression to solve the z-transform of causal periodic signals. 6
- C) State and prove any 6 properties of DTFT. 6

Q5. Solve Any Two of the following.

- A) For $H(z) = 1 + 2z^{-1} - z^{-2} + 3z^{-3} + 3z^{-4} - z^{-5} + 2z^{-6} + z^{-7}$ 6
 Derive $h(n)$ and draw the linear phase FIR implementation.
- B) Compare impulse invariance method and Bilinear transformation method. 6
- C) Explain Gibbs phenomenon. 6

Q6. Solve Any Two of the following.

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| A) | Derive the equation for the order of the low pass Butterworth analog filter. | 6 |
| B) | Explain the operations: Up sampling, Down sampling. | 6 |
| C) | Discuss the applications of Multirate signal processing. | 6 |

***** End *****